Maximum Facts, Minimum Budget

Collection plan for QueryBook using people, carts and the Ray-Ban Meta Gen 2 with Meta's developer toolkit

October 2026 · Recommendations only — no change to QueryBook's design. All prices are rough US estimates; check current prices before buying.

1. The rule that drives everything

Maximize **closed questions per dollar**, not raw volume. A fact counts only when it serves a chosen purpose and a knowledge level. Cost per closed question is measured weekly and decides where the money goes.

Read before you sense. Free, authoritative sources can be ingested at almost no cost:

- traffic codes, US federal sign standards and crash records;
- physics and units tables;
- open datasets such as Wikidata (CC0).

Sensors cost far more per fact, so spend them only on what cannot be read: real scenes, real motion, touch and the local situation.

2. Principles this plan applies

- **Levels:** knowledge is built in levels — perception (L1), concepts (L2), behaviour and rules (L3), situations (L4), prediction (L5), advice (L6), review (L7). Each level rests only on the one below.
- **Purpose first:** nothing is collected unless a chosen purpose needs it. Pilot purposes: P1 “What is this, and where did I see it?” and P2 “Cross the street safely”; later P3 indoor movement and P5 home and kitchen hazards.
- **Return on investment:** foundations first, because everything above reuses them; durable knowledge before fast-changing knowledge; “good enough for the purpose” before fine detail.
- **Questions as a map:** every level records the questions it raises; the open-question record steers the next collection.
- **Directed collection:** QueryBook's harvesting asks for the exact view, sound or touch it needs, instead of recording everything.

3. Who collects what

Knowledge	Cheapest suitable collector	Why
Rules, definitions, physics, statistics (L2–L3)	**Documents and open data**	Close to free, lasting, high reuse
Fixed places — a crossing, a room, day and night (L1, L4)	**Cart with a camera mast**	Steady, repeatable, can run unattended
Hazard scenarios (P2)	**Cart**	No risk to people
Touch — weight, hardness, heat,	**Cart with an arm**	Repeatable; safe with no-

Knowledge	Cheapest suitable collector	Why
grip (L2)		contact heat sensing first
Daily life, the human viewpoint, directed look requests	**Human wearers with the glasses**	Only people give the real-product view
Advice and warnings (L6)	**Human wearers**	Only people can show whether advice is understood

4. Equipment by budget tier

Tier 1 — Starter (about $1,000–1,500)

Item	Approx. cost
Ray-Ban Meta Gen 2, one pair	~$380
Android phone for the toolkit app (or an existing phone)	$0–400
Human-pushed cart: garden or utility wagon, a 1.6 m camera mast, an old phone or webcam as the camera, a power bank	~$150–300
Pan-tilt kit (servos) plus a small single-board computer (e.g. Raspberry Pi), so QueryBook can aim the camera	~$150
1 TB USB SSD (backup mirror)	~$80
Do-it-yourself vibrating wristband (ESP32-class board plus two vibration motors)	~$20–40
Secure link: WireGuard or Tailscale (free tier) and home Wi-Fi	$0
Software: QueryBook, MediaPipe, open-source tools	$0

A cart pushed by a person is the cheapest “robot” there is: a steady, repeatable, eye-height view, with no motors needed to start.

Tier 2 — only after Tier 1 results justify it (add about $800–1,500)

Item	Approx. cost
Low-cost depth camera (e.g. Luxonis OAK-D Lite class)	~$150–300
Open-source low-cost robot arm (e.g. the SO-101 used by Hugging Face's LeRobot project)	~$150–300 in parts
Touch sensing: weighing sensors for force and weight, an infrared thermometer, a contact microphone	~$50
One high-detail touch sensor (DIGIT class)	a few hundred dollars
Motor kit for the cart, so it can run unattended at a fixed site	~$300–500

Tier 3 — only if the measured return says so

- A second pair of glasses for a second wearer.

- A robot dog, only if the routes need kerbs or rough ground. A humanoid robot is not worth it yet for this purpose.

People

Start with yourself plus one or two consenting volunteers. If wearers are paid, budget by the hour and spend wearer time only on human-only work (the last two rows of section 3).

5. Schedule (12 weeks)

Weeks	Work	Cost	Facts expected
1–3	**Read first:** ingest the free sources for P1 and P2 — traffic codes, crash records, physics, object properties	~$0	Most facts per dollar in the plan; the L2–L3 foundation
1–2	Buy Tier 1; test the glasses (stream quality, total delay, endurance) mounted on the cart's mast	Tier 1	L1 measurements of the glasses
3–6	**Cart at fixed sites:** a chosen crossing and a room, day and night; the pan-tilt aimed by QueryBook's look requests	Low (your time)	L1 and L4 scene facts; P2 recordings for replay testing
4–8	**Human wearers:** directed look requests in daily life (P1), 1–2 hours a day each	Wearer time	Human-view L1 facts; confirmations and corrections
7–8	**Review:** cost per closed question for each collector	—	Decision on Tier 2
9–12	Tier 2 if justified: arm and touch sensing for P5; motorized cart at a fixed site; replay tests of P2; wristband against spoken warnings	Tier 2	Measured L2 touch facts; L5 and L6 test results

6. How look requests work

1. The open-question record raises a need (for example: “cannot tell a kerb from a crossing — need views of crossings at night”).
2. QueryBook issues a request: the object, the view needed (angle, distance, lighting), the reason, the priority and an expiry time.
3. The cart's pan-tilt is aimed directly; a human wearer may accept or decline, and is guided by voice through the glasses' speakers (“a little left… hold still… got it”).
4. The request completes only when the object is identified and held in view from the angles, distances and lighting asked for.
5. The capture is filed at the right level and closes the question — or opens a new one.

7. Data link

- Path: glasses → (Bluetooth) → phone → (Wi-Fi or mobile data) → the computer running QueryBook. Carts connect over Wi-Fi.

- **Small live traffic** — requests, what was seen, confirmations — is sent immediately.
- **Video and audio** are stored on the device and uploaded later, by default on Wi-Fi; deleted only after the computer confirms receipt by checksum.
- Encrypted end to end through a private VPN (WireGuard or Tailscale). QueryBook's port is never opened to the internet.
- Each sense — video, audio, touch — is kept in its own stream on one shared timeline. The USB SSD serves as the backup mirror.

8. How the budget is protected

- **Every collector earns its keep:** each week compare cost per closed question for documents, cart, wearers and arm; move effort and money to whichever is cheapest for the open questions; stop expanding anything not being drawn on.
- **Directed, not continuous:** carts and wearers capture in full only when a request is open; otherwise they log only what was seen. This saves storage, data and review time.
- **Collect once, reuse many times:** recorded scenes go into the replay library and are re-run after every improvement, at no extra collection cost.
- **Buy in steps:** Tier 2 and Tier 3 purchases happen only when the numbers from the tier before show a payoff.

9. Safety, consent and privacy

- Wearers give informed consent, may decline any request and can stop at any time; quiet hours and a single “stop all requests” command.
- No requests while a wearer is crossing a road, driving, cycling or on stairs. Hazards are tested with carts or simulations, never aimed at people.
- No identification of bystanders; no look requests aimed at a particular person without that person's agreement.
- Touch: never touch people; robots measure heat without contact first; strict force limits; touch only with permission (not shop goods or other people's property).
- Recording laws differ by place; some places ban camera glasses. If this is research, have an ethics board approve the protocol first.
- Check Meta's developer terms (research and commercial use, where data goes) and keep the design device-independent.

10. What to measure weekly

Measure	Why it matters
Questions closed, per collector	Shows which collector is productive
Cost per closed question (equipment spread over the weeks, plus people's hours)	The main decision number
Facts used in at least one conclusion, against facts stored	Exposes wasted harvesting
Purposes completed (P1, then P2)	Progress measured by capability, not volume

11. Bottom line

About $1,000–1,500 to start. The largest single source of facts — free documents and open data — costs almost nothing. The cart provides cheap round-the-clock coverage and the safe testing; people supply only what only people can. Further spending follows the measured return.